

## Chapter 2 Overview

| 2.1 Vectors in n-Space                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| Learning Outcomes                      | <ul style="list-style-type: none"> <li>• What is an <b>n-vector</b>?</li> <li>• What are vector addition, subtraction and scalar multiplication?</li> <li>• What is an <b>n-space <math>\mathbf{R}^n</math></b>?</li> <li>• What is dot product?</li> <li>• How to find length (norm), distance and angles in <b><math>\mathbf{R}^n</math></b>?</li> <li>• What are orthogonal vectors?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Key Points                             | <ul style="list-style-type: none"> <li>• 2-vectors and 3-vectors and their operations can be represented geometrically in terms of arrows in the xy-plane and xyz-space respectively.</li> <li>• An n-vector can be regarded as <math>1 \times n</math> matrix or <math>n \times 1</math> matrix.</li> <li>• If a linear system has n variables, then its solutions can be regarded as n-vectors, and the solution set is a subset of <b><math>\mathbf{R}^n</math></b>.</li> <li>• Length, distance and angles on n-vectors can be computed in terms of dot product.</li> <li>• Two vectors are orthogonal to each other if their dot product is 0.</li> </ul>                                                                                                                                                                                                                                                                                 |
| 2.2 Linear Combination and Linear Span |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Learning Outcomes                      | <ul style="list-style-type: none"> <li>• What is a <b>linear combination</b>?</li> <li>• How to determine whether a vector is a linear combination of a given set of vectors?</li> <li>• How to express a vector as a linear combination?</li> <li>• What are standard basis vectors?</li> <li>• What is a <b>linear span</b>?</li> <li>• How to determine whether a vector belong to a linear span?</li> <li>• What are the geometrical representation of linear spans of one and two vectors in <b><math>\mathbf{R}^2</math></b> and <b><math>\mathbf{R}^3</math></b>?</li> </ul>                                                                                                                                                                                                                                                                                                                                                            |
| Key Points                             | <ul style="list-style-type: none"> <li>• A linear combination of a set of n-vector is itself an n-vector</li> <li>• A linear system can be expressed in vector equation form.</li> <li>• A vector may be expressed as a linear combination of a fixed set of vectors in more than one way.</li> <li>• Every vector in <b><math>\mathbf{R}^n</math></b> can be expressed as a linear combination of its standard basis vectors.</li> <li>• The linear span of a given set of vectors is the collection of all possible linear combinations of that set of vectors.</li> <li>• The linear span of a single vector in <b><math>\mathbf{R}^2</math></b> and <b><math>\mathbf{R}^3</math></b> is represented by a line that pass through the origin.</li> <li>• The linear span of two non-parallel vectors in <b><math>\mathbf{R}^2</math></b> and <b><math>\mathbf{R}^3</math></b> is represented by a plane that contains the origin.</li> </ul> |

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|                   | <b>2.3 Subspaces</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Learning          | <ul style="list-style-type: none"> <li>What is <b>row space/column space/nullspace</b> of a matrix?</li> <li>What is a <b>vector space</b> and a <b>subspace</b>?</li> <li>What is a <b>solution space</b> of a linear system?</li> <li>What are the closure properties of a subspace?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Key Points        | <ul style="list-style-type: none"> <li>Row space of an <math>m \times n</math> matrix <math>\mathbf{A} = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m\}</math> where <math>\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m</math> are the <math>m</math> rows of <math>\mathbf{A}</math>.</li> <li>Column space of an <math>m \times n</math> matrix <math>\mathbf{A} = \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}</math> where <math>\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n</math> are the <math>n</math> columns of <math>\mathbf{A}</math>.</li> <li>Nullspace of <math>\mathbf{A} =</math> the solution space of <math>\mathbf{Ax} = \mathbf{0}</math></li> <li>The set of all solutions of a homogeneous system with <math>n</math> variables is a subspace of <math>\mathbf{R}^n</math>.</li> <li>A subspace of <math>\mathbf{R}^n</math> can be expressed in <u>linear span form</u> <math>\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}</math></li> <li>A subspace of <math>\mathbf{R}^n</math> satisfies the closure properties under vector addition and scalar multiplication.</li> <li>(In this course), a vector space refers to <math>\mathbf{R}^n</math> (for any <math>n</math>) or a subspace of <math>\mathbf{R}^n</math>.</li> </ul>      |
|                   | <b>2.4 Linear Independence</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Learning Outcomes | <ul style="list-style-type: none"> <li>What is a "redundant vector" in a linear span?</li> <li>What is a <b>linearly independent/dependent</b> set?</li> <li>How to show that a set is linearly (in)dependent?</li> <li>What is the relation between linearly (in)dependence and determinant?</li> <li>What is the relation between linear dependence and redundancy?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Key Points        | <ul style="list-style-type: none"> <li><math>\mathbf{u}</math> and <math>\mathbf{v}</math> are scalar multiples of each other, then they are linearly dependent.</li> <li>A set of vectors <math>\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k</math> is linearly dependent if at least one of the vectors <math>\mathbf{u}_i</math> is a linear combination of the other vectors.</li> <li>A set of <math>n</math> vectors <math>\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n</math> in <math>\mathbf{R}^n</math> is linearly independent if <math>\det(\mathbf{A}) \neq 0</math>, where <math>\mathbf{A}</math> is the <math>n \times n</math> matrix with <math>\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n</math> as its columns.</li> <li>Two vectors <math>\mathbf{u}, \mathbf{v}</math> in <math>\mathbf{R}^2</math> are linearly independent <math>\Leftrightarrow \text{span}\{\mathbf{u}, \mathbf{v}\} = \mathbf{R}^2</math></li> <li>Three vectors <math>\mathbf{u}, \mathbf{v}, \mathbf{w}</math> in <math>\mathbf{R}^3</math> are linearly independent <math>\Leftrightarrow \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \mathbf{R}^3</math></li> <li>If a set of vectors in <math>\mathbf{R}^n</math> has more than <math>n</math> elements, then the set is linearly dependent.</li> </ul> |
|                   | <b>2.5 Basis and Dimension</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Learning Outcomes | <ul style="list-style-type: none"> <li>What is a <b>basis</b> and <b>dimension</b> for a vector space?</li> <li>How to show that a set is a basis for <math>\mathbf{R}^n</math>?</li> <li>How to find bases for row /column/ nullspaces for matrices?</li> <li>How to find a basis for a linear span?</li> <li>How to find a basis and dimension for solution space?</li> <li>What is the nullity of a matrix?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Key Points        | <ul style="list-style-type: none"> <li>The standard basis vectors form a basis for <math>\mathbf{R}^n</math> (for all <math>n</math>)</li> <li>A (non-zero) vector space has many different bases.</li> <li>All bases for the same vector space <math>V</math> has the same number of vectors (called the dimension of <math>V</math>).</li> <li>If <math>S</math> is a set of <math>n</math> linearly independent <math>n</math>-vectors, then <math>S</math> is a basis for <math>\mathbf{R}^n</math>.</li> <li>For a homogeneous system, <math>\dim(\text{solution space}) =</math> number of parameters in the general solution</li> <li><math>\dim(\text{row space of } \mathbf{A}) = \dim(\text{column space of } \mathbf{A}) = \text{rank}(\mathbf{A})</math></li> <li><math>\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) =</math> number of columns of <math>\mathbf{A}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  |

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|                   | <b>2.6 Coordinate Vectors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Learning Outcomes | <ul style="list-style-type: none"> <li>What are <b>coordinate vectors</b> relative to a basis?</li> <li>How to compute coordinate vectors relative to a given basis?</li> <li>How to use a basis to represent a coordinate system?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Key Points        | <ul style="list-style-type: none"> <li>Every vector in a vector space can be expressed as linear combination of a basis in a unique way</li> <li>The coordinate vectors of a fixed (same) vector relative to different bases are different.</li> <li>A basis for a vector space <math>V</math> is a building block for <math>V</math> with the smallest number of vectors</li> <li>A basis for <math>V</math> is a "unit of measurement" for vectors in <math>V</math>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                   | <b>2.7 Projection and Linear Approximation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Learning Outcomes | <ul style="list-style-type: none"> <li>What is the <b>projection</b> of a vector onto a subspace?</li> <li>What is a <b>Least Squares solution</b>?</li> <li>How to find the best approximate solution to inconsistent system?</li> <li>How to use least squares solutions to find projection of a vector onto a certain subspace?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Key Points        | <ul style="list-style-type: none"> <li>If <math>\mathbf{p}</math> is the projection of <math>\mathbf{u}</math> onto a subspace <math>V</math>, then <math>\mathbf{u} - \mathbf{p}</math> is orthogonal to <math>V</math></li> <li>If <math>\mathbf{p}</math> is the projection of <math>\mathbf{u}</math> onto a subspace <math>V</math>, then <math>\ \mathbf{u} - \mathbf{p}\  \leq \ \mathbf{u} - \mathbf{v}\ </math> for any vector <math>\mathbf{v}</math> in <math>V</math>. We say <math>\mathbf{p}</math> is the best approximation of <math>\mathbf{u}</math> in <math>V</math>.</li> <li>The least squares solutions of <math>\mathbf{Ax} = \mathbf{b}</math> is given by the solutions of <math>\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}</math></li> <li>If <math>\mathbf{u}</math> is a least squares solution of <math>\mathbf{Ax} = \mathbf{b}</math>, then <math>\mathbf{Au} = \mathbf{p}</math> gives the projection of <math>\mathbf{b}</math> onto the column space of <math>\mathbf{A}</math>.</li> </ul> |